

DSA0602 - DATA HANDLING AND VISUALIZATION

Name: B Kirthanaa

Register Number: 192324244

SET 1 — Monthly Sales Data

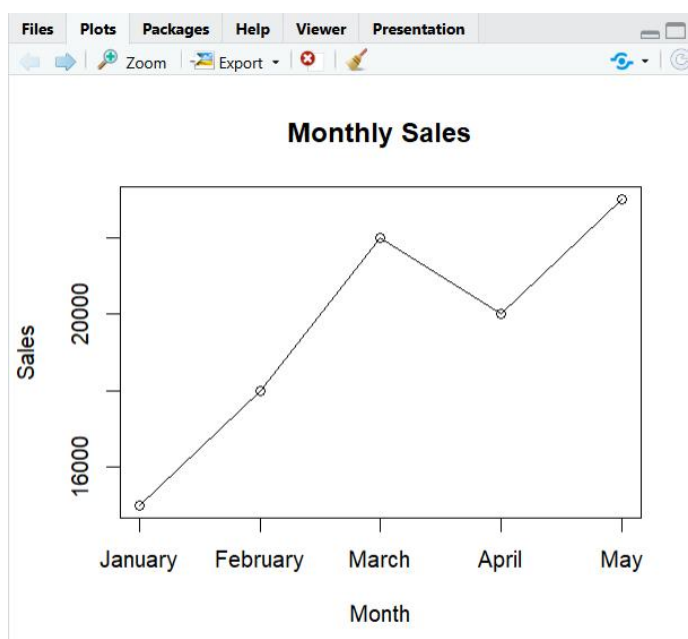
Data

```
month <- c("January", "February", "March", "April", "May")
sales <- c(15000, 18000, 22000, 20000, 23000)
product <- c("Laptop", "Mobile", "Tablet", "Headphone", "Watch")
product_sales <- c(50000, 45000, 30000, 25000, 20000)
advertising <- c(5000, 6000, 7000, 6500, 8000)
df <- data.frame(month, sales, advertising)
```

1. Line Chart — Monthly Sales

```
plot(sales, type="o",
     xaxt="n",
     xlab="Month",
     ylab="Sales ($) ",
     main="Monthly Sales")
axis(1, 1:5, month)
```

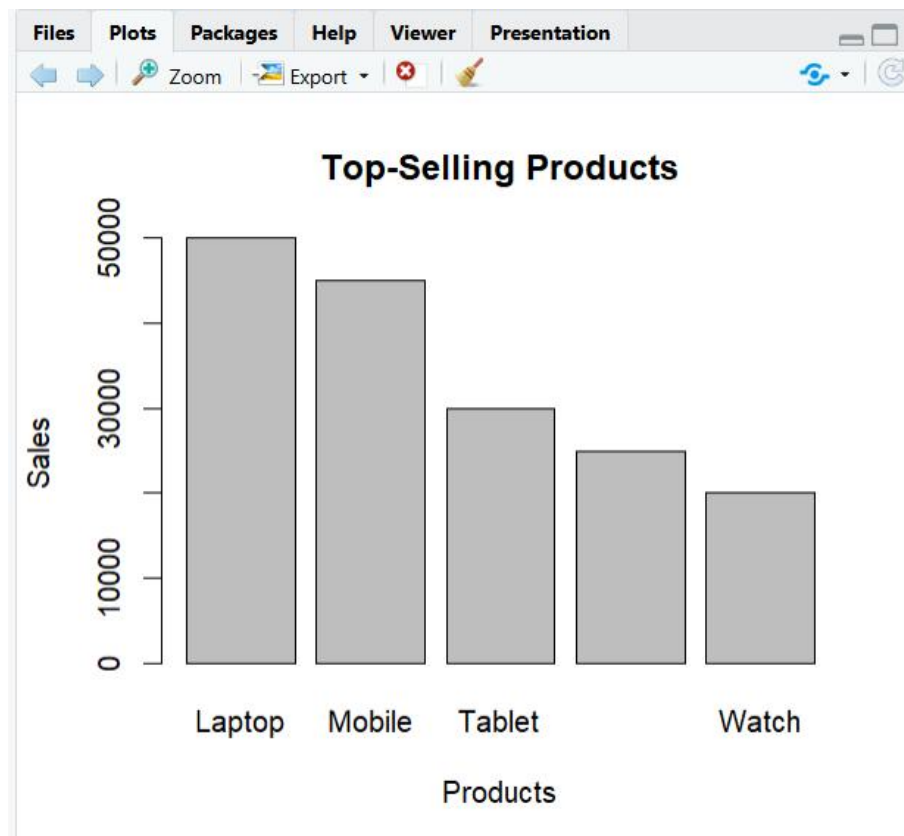
OUTPUT:



2. Bar Chart — Top-Selling Products

```
barplot(product_sales,  
        names.arg=product,  
        xlab="Products",  
        ylab="Sales ($)",  
        main="Top-Selling Products")
```

OUTPUT:

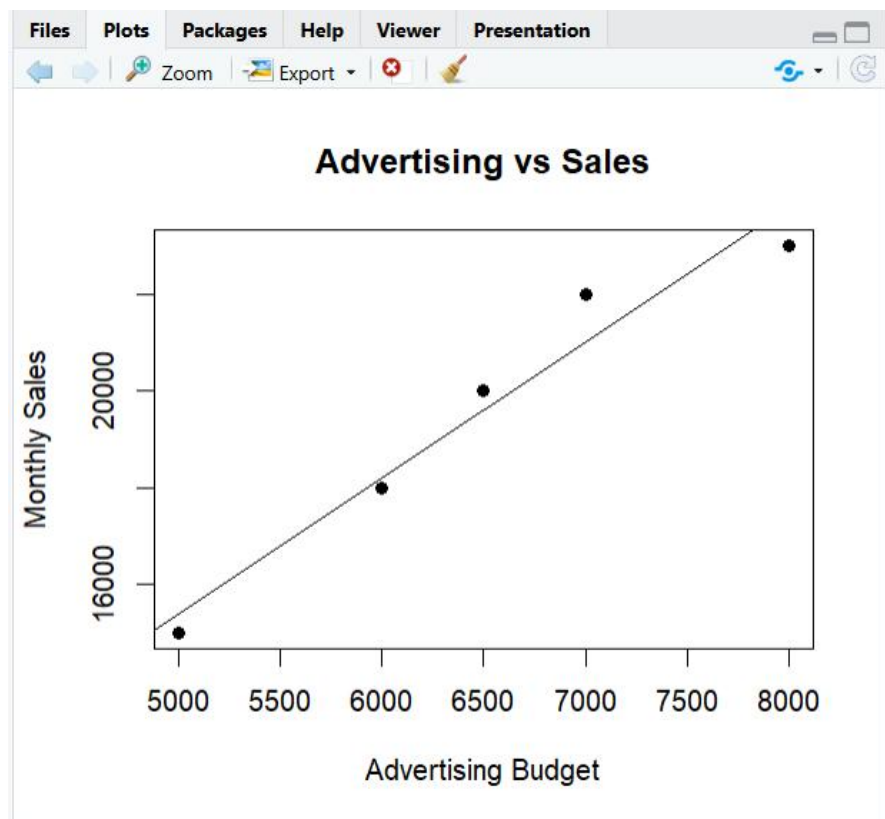


3. Scatter Plot — Advertising vs Sales

```
plot(advertising, sales,  
     xlab="Advertising Budget ($)",  
     ylab="Monthly Sales ($)",  
     main="Advertising Budget vs Sales",  
     pch=19)
```

```
abline(lm(sales ~ advertising), lty=2)
```

OUTPUT:



Insight:

As the advertising budget increases, monthly sales also generally increase. This indicates a positive relationship between advertising and sales.

4. Interactive Dashboard

```
install.packages("shiny")
```

```
library(shiny)
```

```
ui <- fluidPage(
```

```
  titlePanel("Monthly Sales Dashboard"),
```

```
  plotOutput("line"),
```

```
  plotOutput("bar")
```

```
)
```

```
server <- function(input, output) {
```

```
  output$line <- renderPlot({
```

```

plot(sales, type="o",
     xaxt="n",
     xlab="Month",
     ylab="Sales",
     main="Monthly Sales")
axis(1, 1:5, month)
})

```

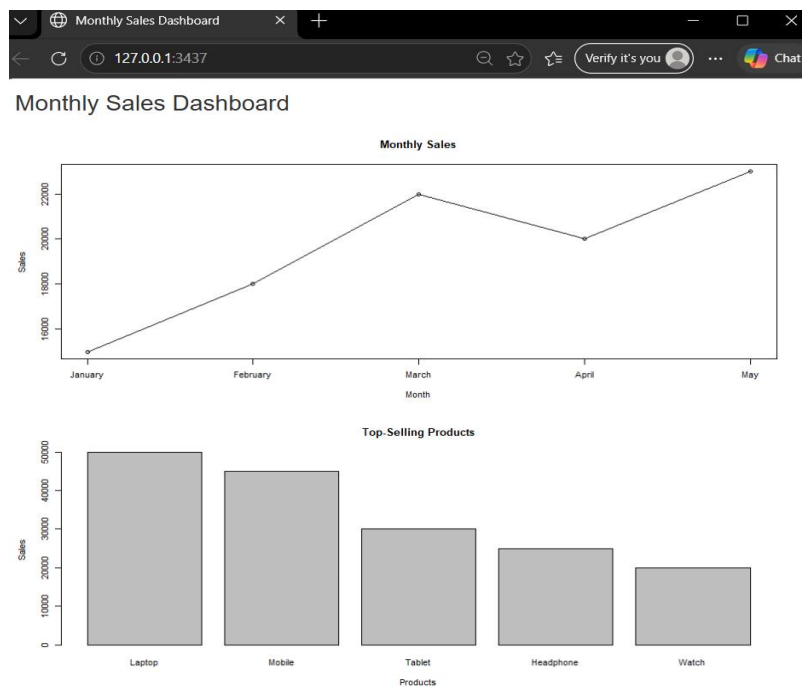
```

output$bar <- renderPlot({
  barplot(product_sales,
          names.arg=product,
          main="Top-Selling Products",
          ylab="Sales")
})
}

```

```
shinyApp(ui, server)
```

OUTPUT: INTERACTIVE-DASHBOARD:



R-STUDIO:

```
SET-1.R x
1 # SET 1: MONTHLY SALES
2
3 # Data
4 month <- c("January", "February", "March", "April", "May")
5 sales <- c(15000, 18000, 22000, 20000, 23000)
6 product <- c("Laptop", "Mobile", "Tablet", "Headphone", "Watch")
7 product_sales <- c(50000, 45000, 30000, 25000, 20000)
8 advertising <- c(5000, 6000, 7000, 6500, 8000)
9
10 # 1. Line Chart
11 plot(sales, type="o", xaxt="n",
12       xlab="Month", ylab="Sales",
13       main="Monthly Sales")
14 axis(1, 1:5, month)
15
16 # 2. Bar Chart
17 barplot(product_sales, names.arg=product,
18         xlab="Products", ylab="Sales",
19         main="Top-Selling Products")
20
21 # 3. Scatter Plot
22 plot(advertising, sales, pch=19,
23       xlab="Advertising Budget",
24       ylab="Monthly Sales",
25       main="Advertising vs Sales")
26 abline(lm(sales ~ advertising))
27
28 # Insight:
29 # Advertising and sales show a positive relationship.
30
31 # 4. Simple Interactive Dashboard using Shiny
32 # Run this part separately after installing shiny:
33 # install.packages("shiny")
34
35 library(shiny)
36
37 ui <- fluidPage(
38   titlePanel("Monthly Sales Dashboard"),
39   plotOutput("line"),
40   plotOutput("bar")
41 )
42
43 server <- function(input, output) {
44   output$line <- renderPlot({
45     plot(sales, type="o", xaxt="n",
46          xlab="Month", ylab="Sales",
47          main="Monthly Sales")
48     axis(1, 1:5, month)
49   })
50
51   output$bar <- renderPlot({
52     barplot(product_sales, names.arg=product,
53            xlab="Products", ylab="Sales",
54            main="Top-Selling Products")
55   })
56 }
57
58 shinyApp(ui, server)
```

28:11 (Top Level) ↕